Uncertain Example Mining Network for Domain Adaptive Segmentation of Remote Sensing Images

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***Abstract*— Domain adaptive segmentation has recently gained more and more attention in the remote sensing field. However, current methods often generate a significant number of uncertain examples, i.e., noisy pseudo-labels, in the target domain, which adversely affects model convergence. To solve this issue, an uncertain example mining network is proposed for domain adaptive segmentation of remote sensing images. Specifically, a novel strategy called multilevel pseudo-label correcting (MPC) is proposed to correct the pseudo-labels in class, pixel, and superpixel levels. In this way, more reliable pseudo-labels can be selected for the subsequent training stage. Furthermore, a noise-robust example mining strategy, termed uncertainty-based valuable example mining (UVEM), is proposed to prioritize confident examples with significant gradients for training effectively. Extensive empirical evaluations on IsprsDA and LoveDA datasets demonstrate that the proposed method outperforms previous approaches, establishing state-of-the-art results in domain adaptive remote sensing image segmentation (RSIS). The code will be available at https://github.com/StuLiu/UemDA.**

***Index Terms*— Domain adaptation, noise label correction, remote sensing image segmentation (RSIS), self-training (ST).**

I. Introduction

SEMANTIC segmentation is a foundational problem in computer vision and holds a pivotal position in the field of remote sensing [1]. In the past decades, extensive efforts have been made to develop all kinds of semantic segmentation methods. Among these methods, deep learning-based segmentation methods have achieved excellent performance in speed and accuracy when the training and testing data belong to the same domain. However, when the testing set is previously unseen out-of-domain data, current deep learning-based segmentation methods suffer from a notable degradation [2], [3], [4].

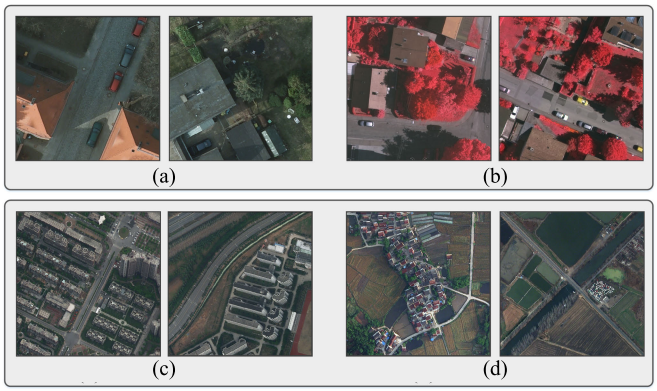


Fig. 1. Illustration of the domain shift. (a) and (b) IsprsDA datasets obtained in different imaging locations by different sensors. (c) and (d) LoveDA datasets gained in different locations.

The performance degradation is attributed to distribution variation between domains, also known as domain gaps. These domain gaps are usually caused by the different imaging locations and sensors [5]. As shown in Fig. 1(a)–(d), there are great differences in the object style and size across different imaging locations. The buildings in the urban area have a disparate style than the ones in the rural area, and the roads in the urban area are wider than the ones in the rural area. Furthermore, various imaging sensors can lead to spectral variations across domains. As shown in Fig. 1(a) and (b), the image value in the red channel of the Vaihingen (IRRG) is greater than the ones in Potsdam (RGB). In this work, we merely focus on solving the problem of the domain gaps caused by discrepant imaging locations and imaging sensors.

The unsupervised domain adaptive (UDA) segmentation technique is dedicated to addressing the intractable problem of remote sensing image segmentation (RSIS) without any target domain data during the training stage. In recent years, self-training (ST)-based methods have gained popularity in the field of UDA due to their stable training process and significant performance improvement. However, the ST procedure often leads to generating a considerable number of false pseudo-labels or so-called noisy labels on the target domain, which can have a detrimental impact on the convergence

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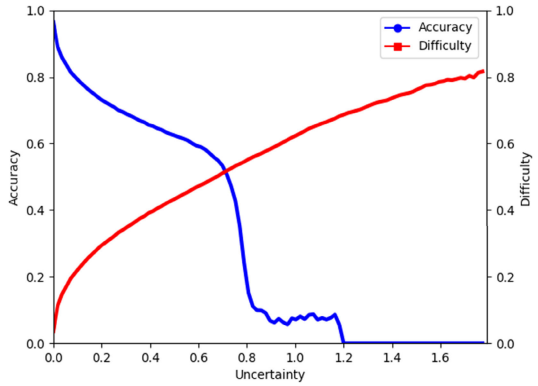


Fig. 2. Relationship of the difficulty and the accuracy with respect to the uncertainty.

of the model. Furthermore, as depicted in Fig. 2, the noisy labels exhibit an uneven distribution across different difficulty levels. The examples with a larger gap between prediction and ground truth may be more prone to noisy pseudo-labels than others, which poses a significant challenge for the hard example mining algorithm. In the remote sensing field, current UDA methods [2], [3], [5], [6], [7] hardly consider the noisy pseudo-labels in the target domain, which primarily aims to mitigate the adverse effects caused by these uncertain examples during the ST process.

In the computer vision field, some noisy pseudo-label correcting methods have been developed, which can be classified into two groups: noisy label correcting methods and loss weight adjusting methods. On the one hand, researchers make efforts to generate more accurate pseudo-labels by maintaining prototypes [8], [9] or well-designed confidence thresholds [10], [11]. However, these methods ignore the output probability as well as the local spatial consistency of the input data. On the other hand, adjusting loss weights shows promise in alleviating the negative effect of the false labels by assigning different loss weights for the training examples [12]. In [13], the high-uncertainty examples are assigned with a zero weight. In [14], the outlier examples are assigned with a low weight to prevent these examples from participating in the training process. Nevertheless, it is difficult to assign the optimal weight for the uncertain examples in the target domain due to the conflict between high accuracy and the large magnitude of the gradient.

To solve these issues, this article proposes an uncertain example mining network for domain adaptive segmentation of remote sensing images. First, the multilevel correction method is designed for noisy pseudo-label removal, which aims to enhance the accuracy of pseudo-labels by adjusting their probabilities from multiple levels, including feature consistency (class-level), probability consistency (pixel-level), and spatial consistency (superpixel-level). Furthermore, this work proposes an uncertainty-based valuable example mining (UVEM) method to emphasize the significance of valuable examples while filtering out erroneous pseudo-labels during the training stage. Through extensive empirical evaluations on benchmark datasets, our approach achieves superior performance than previous UDA methods for RSIS. In summary, the main contributions of this article can be summarized as follows.

1) An uncertain example mining network is proposed to address the challenges inherent in ST-based domain adaptation for RSIS.
2) A novel multilevel pseudo-label correcting (MPC) approach is proposed. It aims to enhance the accuracy of pseudo-labels by adjusting the soft pseudo-labels in class, pixel, and superpixel levels.
3) An UVEM algorithm is proposed. It emphasizes the significance of valuable examples while filtering out erroneous pseudo-labels during the training stage.
4) Extensive experiments on commonly adopted datasets demonstrate the effectiveness of the proposed MPC and UVEM. Compared with other domain adaptive segmentation methods, our uncertain example mining network achieves outstanding segmentation performance.

II. Related Work

In this section, a comprehensive overview of the previous deep UDA approaches and their applications in RSIS are provided. Subsequently, the advancements in learning from noisy labels are given. Finally, the current research efforts on the topic of example mining are reviewed.

A. UDA Methods

UDA aims to narrow the domain discrepancy between the source and target domains [15], [16]. It seeks a generalized model by aligning the input image distribution or feature distribution between the source and target domains. Among the fields of domain adaptation, UDA is a typical task that works without any labeled data in the target domain. Recent UDA methods are based on deep neural networks yielding superior performance than traditional methods due to more expressive feature extraction and the end-to-end requirements in specific applications. The deep UDA methods can be classified into discrepancy-based [17], [18], adversarial training (AT) [19], [20], [21], contrastive learning (CL) [6], [22], [23], ST methods [10], [24], [25], [26], and mixing (MIX) [25], [27]. First, discrepancy-based methods aim to mitigate dissimilarities between two domains by aligning their respective global data distributions [16], but ignoring the distribution discrepancy on class level and pixel level. Second, AT has become increasingly popular for aligning representations of the source and target domains. Nevertheless, the training process of the AT is unstable. In addition, CL clusters deep features for cross-domain pixel pairs with the same category, while pushing apart those with different classes in the feature space. Moreover, ST is another commonly used technique in domain adaptation, which adapts the model (both encoder and classifier) by leveraging pseudo-labels and offers stable training processes. Finally, MIX randomly generates domain-mixed images by segmenting images from both the source and target domains, then trains the model by the source and the domain-mixed images. Most of these approaches rely on selecting high-confidence labels as pseudo-labels for training, enabling

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the exploration of target domain information. However, generating noise-free pseudo-labels still poses a challenge.

Various UDA techniques have been explored in RSIS. Na et al. [28] propose an adversarial transfer attack scheme that aligns the distributions of the images of source and target domains. Chen et al. [29] propose a region-aware and category-aware GAN to consider the diversities presented in different regions within a specific image, as well as the distribution at the category level. Zhou et al. [7] take DSM data into account to improve domain adaptation performance. Xi et al. [3] employ a curriculum learning approach to align the feature distribution across domains. This approach first aligns the sample relationships at the sample level. It then utilizes KL divergence to reduce the distances between class prototypes at the category level. Finally, pixel-level pseudo labels are generated for ST. Wu et al. [6] propose a correlation alignment scheme to minimize the distance between the features of the source domain and the ones of the target domain on the category level. Overall, UDA has great potential in addressing the domain shift challenges in RSIS. Nevertheless, the application of ST-based techniques in RSIS remains a challenging problem that requires further research and exploration.

B. Learning From Noisy Labels

Noisy labels are widespread in supervised machine-learning tasks. Effectively removing noise is a crucial element in supervised learning for obtaining dependable models from noisy data [30], [31]. Several label-denoising methods are proposed [9] to correct the noisy labels from the source domain. Nevertheless, the target domain exhibits a substantial presence of erroneous pseudo-labels. These noisy pseudo-labels significantly undermine the performance of the model. To deal with this issue, several pseudo-label denoising techniques are proposed, including noise-robust architecture [32], [33], robust loss function [34], [35], pseudo-label correction [8], [9], and reweighting strategy [3], [13]. First, designing an intrinsic noise-robust loss function is one of the commonly employed approaches, such as symmetric cross-entropy loss [35] and generalized cross-entropy loss [34]. Second, adjusting the false pseudo-labels to the true ones is one of the promising approaches. The distances between class prototypes and deep features are typically computed to rectify the predicted pseudo-labels [8], [9]. However, these correction techniques solely estimate the reliability of pseudo-labels from a single perspective while ignoring the learned fresh knowledge and the local spatial consistency. Biswas and Tešić [36] preserve a support set to filter out false pseudo-labels whose features do not align with the support samples. It corrects the hard labels (IDs) through a voting approach at pixel level and gains incredible precision improvement. In contrast, our approach corrects the soft labels (probabilities) through a reweighting process at the class, pixel, and superpixel levels. Finally, Xi et al. [3] propose an adaptive hierarchical pseudo label weighting approach that assigns a dynamic loss weight for each pixel. This method maintains two confidence thresholds over the dataset allowing more hard samples to participate in the training process. However, the pseudo-labels with high confidence are still probably to be noisy labels. In this work, an additional evaluating factor, uncertainty, is introduced to assist in more accurate weight assignment.

C. Example Mining

Previous example mining methods focus on addressing the imbalance of easy and hard examples, which help the model converge better. However, there is a scarcity of research that utilizes hard example mining algorithms [37], [38] for domain adaptation in the context of ST due to the challenges posed by the noises associated with hard examples. These approaches often lead to inferior performance compared to the baseline methods. Gradient harmonization mining (GHM) [14] utilizes the density of the gradient norm to suppress the extremely hard examples. However, the GHM in the context of ST failed since the gradient density is unsuitable for computing loss weight under the high noise situation. Rizve et al. [13] assigns the examples that have high uncertainty a zero weight to avoid these examples participating in the training process. Nevertheless, the weight assignment for examples of the target domain still needs to be developed, since the trade-off between high accuracy and large gradient of pseudo-label examples is not investigated.

III. Methods

A. Method Overview

An overview of the proposed method is illustrated in Fig. 3. The UDA framework is constructed based on the ST architecture which contains two segmentation networks, a teacher model and a Siamese student model. The mission of the teacher model is to provide a stable class probability and generate pseudo-labels for the target domain, while the purpose of the student model which is trained on both the source and target domains is to learn fresh knowledge and update the teacher model. However, the pseudo-labels are noisy leading to the challenge of training a robust model. To alleviate this problem, the MPC module is proposed to correct the pseudo-labels from multiple levels. Furthermore, the UVEM strategy is proposed to emphasize the most valuable examples and suppress the noisy ones making the model converge better. Because of the proposed MPC module and UVME strategy, our method is robust to the noisy pseudo-labels.

B. ST for UDA

Given the images $\mathcal{X}_S = \{\mathbf{X}_S^i\}_{i=1}^{N_S}$ with labels $\mathcal{Y}_S = \{\mathbf{Y}_S^i\}_{i=1}^{N_S}$ for source domain and the images $\mathcal{X}_T = \{\mathbf{X}_T^i\}_{i=1}^{N_T}$ without labels for the target domain, we focus on training an image segmentation network f_θ that gains considerable performance on the target domain. In a supervised manner, the network trained on the source domain usually results in poor performance when they test on the target domain due to the domain shift. The cross-entropy loss is utilized to train the model on the source domain. It can be formulated as follows:

$$\mathcal{L}_S = -\sum_{i=1}^{B}\sum_{j=1}^{H\times W}\sum_{c=1}^{C}\mathbf{Y}_S^{i,j,c}\log\mathbf{P}_S^{i,j,c} \tag{1}$$

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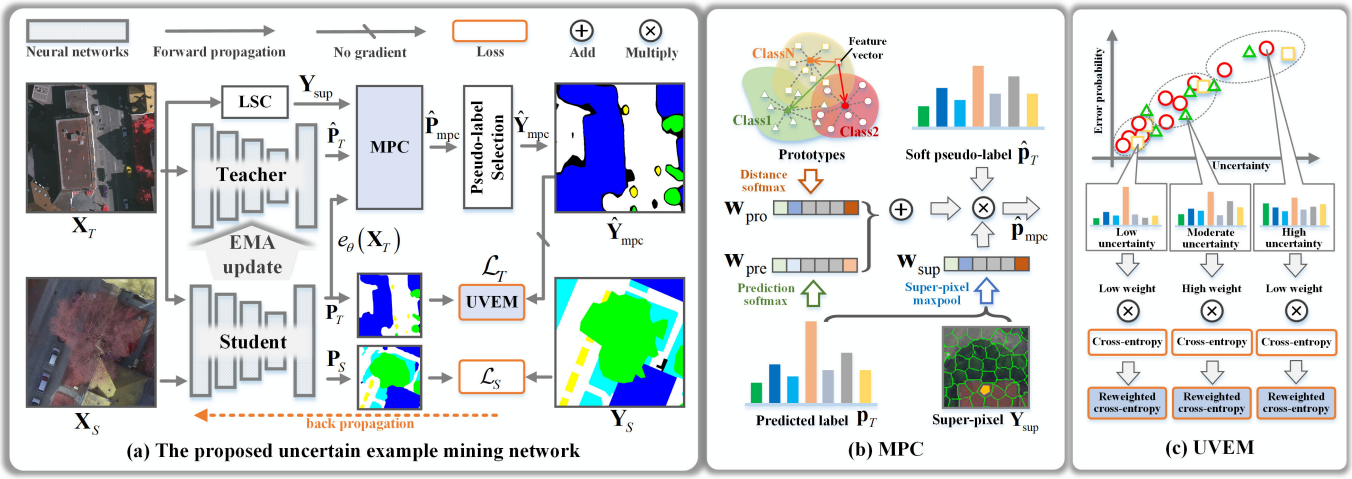


Fig. 3. Overview of the proposed method. (a) ST procedure enhanced by MPC and UVEM for UDA. (b) MPC algorithm. (c) Illustration of the proposed UVEM method. LSC indicates the linear spectral clustering algorithm for super-pixel segmentation.

where B is the number of input images within a mini-batch. H and W indicate the height and the width of the input image $\mathbf{X}_S$. C denotes the number of classes of the datasets. $\mathbf{P}_S$ represents the predicted probabilities, $\mathbf{P}_S = f_\theta(\mathbf{X}_S)$. f_θ is the student network for RSIS.

To tackle the issue of domain shift, ST offers a solution where a teacher model [39] trained on the source domain generates pseudo-labels, $\hat{Y}_T$, for target domain images. Typically, the pseudo-label selection function can be expressed as follows:

$$\hat{Y}_T^{i,j,c} = \left[c = \arg\max_{c'} \hat{P}_T^{i,j,c'} \right] \cdot \left[\hat{P}_T^{i,j,c} \geq \tau^{i,c} \right] \tag{2}$$

where $[\cdot]$ indicates the Iverson bracket. $\hat{P}_T$ represents the predictions of the teacher model $f_{\hat{\theta}}$, that is $\hat{P}_T = f_{\hat{\theta}}(\mathbf{X}_T)$. $\tau^{i,c}$ denotes the confidence threshold for the cth class of the ith image to filter the high-confidence examples. In this work, a simple dynamic threshold adjustment policy is employed as follows:

$$\tau^{i,c} = \max\left(\gamma \cdot \max_{1 \leq j' \leq H \times W} \hat{P}^{i,j',c}, \tau_0 \right) \tag{3}$$

where γ and τ_0 indicate the tolerance ratio and fixed minimum confidence, respectively. In simpler terms, the confidence of the selected pseudo-labels must be higher than both the tolerated confidence threshold $\gamma \cdot \max_{1 \leq j' \leq H \times W} \hat{\mathbf{p}}^{i,j',c}$ and τ_0.

These pseudo-labels $\hat{Y}_T$ are subsequently used to guide the model training by the following loss function:

$$\mathcal{L}_T = -\sum_{i=1}^{B} \sum_{j=1}^{H \times W} \sum_{c=1}^{C} \hat{Y}_T^{i,j,c} \log \mathbf{P}_T^{i,j,c}. \tag{4}$$

By combining the cross-entropy losses on source domain and target domain, the ST loss function can be formulated as follows:

$$\mathcal{L}_{\text{seg}} = \mathcal{L}_S + [t > t_0] \cdot \mathcal{L}_T \tag{5}$$

where t represents the current iteration number. t_0 indicates the number of iterations of the first stage. The model only learns from the source domain when the training iteration is smaller than t_0. Moreover, the source domain and target domain data are both used for training when the iteration number is larger than t_0.

In the ST procedure, the weights of the teacher model are updated by the ones of the student model using the exponentially moving average (EMA) algorithm

$$\hat{\theta}_{t+1} = \alpha \hat{\theta}_t + (1 - \alpha)\theta_t \tag{6}$$

where α is the averaging weight which is a float number smaller than 1.0.

C. Multilevel Pseudo-Label Correcting

Although the basic ST scheme can enhance segmentation performance on the target domain to a certain extent, it still encounters the issue of noisy pseudo-labels. Pseudo-label correction approaches offer a potential solution to mitigate this problem. Online and offline pseudo-label correction manners are two commonly employed schemes. Generally, an online pseudo-label correcting manner can effectively leverage real-time learned knowledge compared to the offline manner. In this work, the online correcting manner can be formulated as follows:

$$\hat{Y}_{\text{mpc}} = \xi\left(\hat{P}_{\text{mpc}}\right) \tag{7a}$$

$$\hat{P}_{\text{mpc}} = \text{Norm}\left(\mathbf{W}_{\text{mpc}} \odot \hat{P}_T\right) \tag{7b}$$

where $\hat{Y}_{\text{mpc}}$ represents the one-hot matrix of the selected hard pseudo-label. $\xi(\cdot)$ indicates the pseudo-label selection which is formulated in (2) and (3). $\hat{P}_{\text{mpc}}$ is the corrected soft pseudo-labels. $\odot$ represents the element-wise multiplication of two matrices with the same size. $\mathbf{W}_{\text{mpc}}$ is the correcting weight to adjust the predicted soft pseudo-labels $\hat{P}_T$ of the teacher model. Norm$(\cdot)$ indicates the normalization. The selected pseudo-labels are employed for subsequent ST iterations.

Previous pseudo-label correcting methods only consider a class level, which has limitations in accurately correcting false pseudo-labels. In this work, an MPC method, i.e., class-level, pixel-level, and superpixel-level, is proposed to estimate the confidence of pseudo-labels from multiple perspectives.

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1) Class-Level View: The cluster assumption in semi-supervised learning suggests that examples with similar deep features tend to group together, implying that examples within a specific cluster can be assigned the same semantic category. Therefore, a class-level correction is constructed to estimate the reliability of the pseudo-labels. More specifically, the local prototype of each class within a mini-batch is first calculated by source features

$$\mathbf{v}^c = \frac{\sum_{i=1}^{B}\sum_{j=1}^{H\times W}\left(e_\theta\left(\mathbf{X}_S^i\right)^{j,c}\cdot\left[\hat{Y}_S^{i,j,c}=1\right]\right)}{\sum_{i=1}^{B}\sum_{j=1}^{H\times W}\left[\hat{Y}_S^{i,j,c}=1\right]} \tag{8}$$

where e_θ indicates the encoder of the student model to produce the deep features. Then, the EMA algorithm is used to update the global prototype

$$\hat{\mathbf{v}}_{t+1} = \alpha\hat{\mathbf{v}}_t + (1-\alpha)\mathbf{v}_{t+1} \tag{9}$$

where $\hat{\mathbf{v}}_t$ and $\mathbf{v}_t$ indicates the global prototypes and local prototypes in iteration t, respectively. When the example features are close to the prototype, the example belongs to the class of the prototype. Finally, the correcting weight of the class-level correction is estimated as

$$\mathbf{W}_{\text{pro}}^{i,j,c} = \frac{\exp\left(d\left(e_\theta\left(\mathbf{X}_T^i\right)^{j,c},\hat{v}^c\right)^{-1}\Big/\eta\right)}{\sum_{c'=1}^{C}\exp\left(d\left(e_\theta\left(\mathbf{X}_T^i\right)^{j,c},\hat{v}^{c'}\right)^{-1}\Big/\eta\right)} \tag{10}$$

where η is the softmax temperature which can control the smoothness of the correcting weight. Specifically, the larger η is, the smoother $\mathbf{W}_{\text{pro}}$ are. $d(\cdot)$ indicates the distance function to compute the discrepancy between two vectors. In this work, the distance function is computed based on the Pearson correlation coefficient [40]

$$d(\mathbf{v}_1,\mathbf{v}_2) = \frac{1-\text{corr}(\mathbf{v}_1,\mathbf{v}_2)}{2} \tag{11}$$

where corr$(\cdot)$ represents the Pearson correlation coefficient between two vectors.

2) Pixel-Level Correction: In our ST procedure, the student network is trained on the source and target domain data, while the teacher network is subsequently updated with the EMA algorithm. Considering that the online pseudo-label generation suffers from the trivial solutions [8], the pseudo-labels are updated every Δt iteration. Accordingly, the student network has the potential to make high-confidence predictions that are more accurate than the fixed pseudo-labels after training several iterations. Therefore, the predicted result for each instance obtained by the student network is utilized to correct the weight by using a softmax normalization

$$\mathbf{W}_{\text{pre}}^{i,j,c} = \frac{\exp\left(\mathbf{P}_T^{i,j,c}\Big/\eta\right)}{\sum_{c'=1}^{C}\exp\left(\mathbf{P}_T^{i,j,c'}\Big/\eta\right)} \tag{12}$$

where η is consistent with (10).

3) Superpixel-Level Correction: Both the class-level and pixel-level corrections adjust the soft pseudo-label for each pixel. However, the local spatial consistency of the input image is not considered. To solve this issue, considering that the pixels in a local homogeneous region often belong to the same category, superpixel-level correction is designed to estimate the reliability of the pseudo-labels. Specifically, a super-pixel segmentation method, i.e., linear spectral clustering (LSC) [41], is utilized to generate local homogeneous regions of images in the target domain

$$\mathbf{Y}_{\text{sup}} = \text{LSC}(\mathbf{X}_T) \tag{13}$$

where x_T represents the image in the target domain. $\mathbf{y}_{\text{sup}}$ is the local homogeneous regions $\mathbf{Y}_{\text{sup}} \in \mathbb{N}^{H\times W}$.

Then, a max-pooling operation is performed on probability maps of each local region

$$\mathbf{P}_{\max}^{k,c} = \max_{j\in\left\{j'\left|\mathbf{Y}_{\text{sup}}^{j'}=k\right.\right\}}\mathbf{P}_T^{j,c} \tag{14}$$

where $\mathbf{P}_{\max}^{k,c}$ indicates the probability map of the cth class for the kth local region. $\mathbf{P}_T$ is the predicted probability maps of the student network. Next, $\mathbf{P}_{\max}^{k,c}$ is assigned to all pixels in each local region

$$\mathbf{P}_{\text{sup}}^{j,c} \leftarrow \mathbf{P}_{\max}^{k,c},\ (j,k)\in\left\{\left(j',k'\right)\middle|\mathbf{Y}_{\text{sup}}^{j'}=k'\right\}. \tag{15}$$

Finally, the correcting weight of the superpixel-level correction is calculated by normalizing the $\mathbf{P}_{\text{sup}}^{j,c}$

$$\mathbf{W}_{\text{sup}}^{i,j,c} = \frac{\exp\left(\mathbf{P}_{\text{sup}}^{i,j,c}\Big/\eta\right)}{\sum_{c'=1}^{C}\exp\left(\mathbf{P}_{\text{sup}}^{i,j,c'}\Big/\eta\right)} \tag{16}$$

where η is the smooth factor.

4) Multilevel Correction: We correct the noisy pseudo-labels by combining the adjusting weight from the class, pixel, and super-pixel levels. To inject the local consistency constraint of the super-pixel level into the class and pixel levels, the multiplication operation is utilized. Thus, the final correcting weight can be computed by the following equation:

$$\mathbf{W}_{\text{mpc}}^{c} = \frac{\left(\mathbf{W}_{\text{pro}}^{c}+\mathbf{W}_{\text{pre}}^{c}\right)\cdot\mathbf{W}_{\text{sup}}^{c}}{\max_{1\le c'\le C}\left(\left(\mathbf{W}_{\text{pro}}^{c'}+\mathbf{W}_{\text{pre}}^{c'}\right)\cdot\mathbf{W}_{\text{sup}}^{c'}\right)} \tag{17}$$

where $\mathbf{W}_{\text{pro}}$, $\mathbf{W}_{\text{pre}}$, and $\mathbf{W}_{\text{sup}}$ are the adjusting weight computed by (10), (12), and (16), respectively.

D. Uncertainty-Based Valuable Example Mining

Previous hard example mining methods [14], [37], [38] have made great progress in supervised deep learning. The hard examples are those misclassified examples with large training gradients while the easy examples are those well-classified examples provide little gradient for the model to learn from. The core idea behind hard example mining is that hard examples usually are assigned greater loss weights making the hard examples contribute more than easy examples. However, in the ST, the predicted pseudo-labels in the target domain are

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polluted with noisy labels, especially those hard examples. As shown in Fig. 2, the higher the difficulty of a sample, the more likely it is to be a noisy example. This is the reason that the hard example mining methods in the ST framework cannot work. To alleviate the issue, the UVEM algorithm is proposed to emphasize the significance of valuable examples while filtering out erroneous pseudo-labels during the training phase. To achieve this goal, the definition of the valuable pseudo-label examples is given as follows.

Definition 1 (Valuable Example): A valuable pseudo-labeled example is one with moderate uncertainty. On one hand, an easy example with extremely low uncertainty yields a meager gradient magnitude and is classified as an invaluable example. On the other hand, a hard example with high uncertainty is likely to be erroneous and is also classified as an invaluable example.

The example uncertainty u can be measured using either entropy or the standard distance of the predictions. In this study, the entropy is chosen as uncertainty measure for each example

$$u = -\sum_{c'=1}^{C} \hat{\mathbf{p}}_T^{c'} \cdot \log\left(\hat{\mathbf{p}}_T^{c'}\right), u \in \left[0, \log(C)\right] \tag{18}$$

where $\hat{\mathbf{p}}_T^{c'}$ indicates the soft pseudo-label vector.

To mine the valuable examples, an uncertainty-based valuable weight w_v is computed for each example

$$w_v = \begin{cases} \sqrt[\rho]{1 - \frac{1}{{u_m}^2}(u - u_m)^2}, & 0 \le u < u_m \\ \sqrt[\rho]{1 - \frac{1}{(u_t - u_m)^2}(u - u_m)^2}, & u_m \le u < u_t \\ 0, & u \ge u_t \end{cases} \tag{19}$$

where u_m indicates the uncertainty of the maximum valuable examples. u_t indicates the gated threshold. u_m and u_t satisfy $0 \le u_m \le u_t \le \log(C)$. The importance of easy examples increases as u_m approaches 0, while the importance of hard examples increases as u_m approaches u_t. Examples with uncertainty closer to u_m are deemed more valuable, while examples with uncertainty exceeding u_t are excluded from the training process. ρ is the adjusting factor. The smaller ρ is, the more important the valuable examples are. The valuable weight curve is illustrated in Fig. 4.

Finally, the loss function in the target domain is computed as

$$\mathcal{L}_T = -\sum_{i=1}^{B}\sum_{j=1}^{H\times W}\left(\mathbf{W}_v^{i,j}\sum_{c=1}^{C}\hat{Y}_{\text{mpc}}^{i,j,c}\log \mathbf{P}_T^{i,j,c}\right). \tag{20}$$

By combining (19) and (20), the examples with uncertainties close to u_m will be assigned higher weights. Consequently, these examples will have a greater influence on the optimization of model parameters.

E. Training Pipeline

The training pipeline of the proposed uncertain example mining network is presented in Algorithm. 1. The training procedure contains two stages. In the first stage, the student network f_θ is trained on the source domain by a generally used cross-entropy loss function. In the meantime, the teacher network $f_{\hat{\theta}}$ is updated by the EMA algorithm. In the second stage, f_θ is trained by both of the source and target domains. Specifically, the teacher network is utilized to generate the soft pseudo-labels $\hat{P}_T$ at first. Since $\hat{P}_T$ is too noisy to train the student network well, the proposed MPC module is designed to rectify noisy pseudo-labels. Then, the pseudo-labels selection in (3) is utilized to gain the rectified pseudo-labels $\hat{Y}_T$. Nevertheless, the rectified pseudo-labels $\hat{Y}_T$ still contain some noises. Therefore, the proposed UVEM is introduced to suppress the noisy examples and mine the valuable ones.

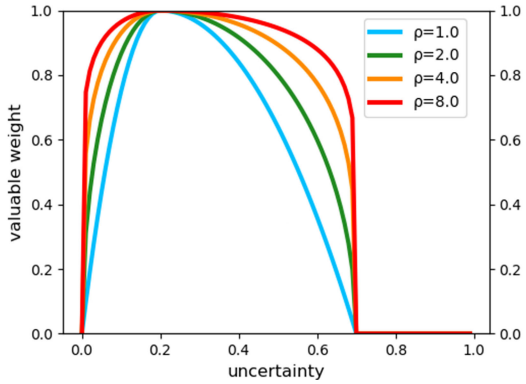


Fig. 4. Illustration of the valuable weight curves under different ρ when $u_m = 0.2$ and $u_t = 0.7$.

Algorithm 1 Training Pipeline

Input:
Source domain datasets $\mathcal{X}_S$ and $\mathcal{Y}_S$, target domain images $\mathcal{X}_T$, segmentation network f_θ, the iteration number of the first stage t_0, the iteration number of the second stage t_1.

Output:
Converged network f_θ.

Initialize student network θ and teacher network $\hat{\theta}$.
/*stage one: training on source domain and aligning features in domain level by Coral loss [18].*/
for $iter = 1$ to t_0 **do**
  $\mathbf{X}_S, \mathbf{Y}_S, \mathbf{X}_T \sim \mathcal{X}_S, \mathcal{Y}_S, \mathcal{X}_T$
  $\mathbf{P}_S \leftarrow f_\theta(\mathbf{X}_S)$
  $\ell_S \leftarrow \mathcal{L}_S(\mathbf{P}_S, \mathbf{Y}_S) + \text{Coral}(e_\theta(\mathbf{X}_S), e_\theta(\mathbf{X}_T))$
  Compute gradient on ℓ_S and optimize θ
  $\hat{\theta} \leftarrow \alpha\hat{\theta} + (1-\alpha)\theta$
end for.
/*stage two: training on source and target domains*/
$\hat{\theta}_{\Delta t} \leftarrow \hat{\theta}$.
for $iter = 1$ to t_1 **do**
  $\mathbf{X}_S, \mathbf{Y}_S, \mathbf{X}_T \sim \mathcal{X}_S, \mathcal{Y}_S, \mathcal{X}_T$
  $\mathbf{P}_S, \mathbf{P}_T \leftarrow f_\theta(\mathbf{X}_S), f_\theta(\mathbf{X}_T)$
  $\hat{P}_T \leftarrow f_{\hat{\theta}_{\Delta t}}(\mathbf{X}_T)$ /*generate soft pseudo-labels*/
  $\mathbf{W}_{\text{mpc}} \leftarrow$ Compute correcting weight by Eq. 17
  $\hat{Y}_{\text{mpc}} \leftarrow \xi(\mathbf{W}_{\text{mpc}} \cdot \hat{\mathbf{P}}_T)$
  $\mathbf{W}_v \leftarrow$ Compute valuable weight matrix by Eq. 19
  $\ell_{\text{seg}} \leftarrow \mathcal{L}_S(\mathbf{P}_S, \mathbf{Y}_S) + \mathcal{L}_T(\mathbf{P}_T, \hat{Y}_{\text{mpc}}, \mathbf{W}_v)$
  Compute gradient on ℓ_{seg} and optimize θ
  $\hat{\theta} \leftarrow \alpha\hat{\theta} + (1-\alpha)\theta$.
  if $iter\%\Delta t = 0$
    $\hat{\theta}_{\Delta t} \leftarrow \hat{\theta}$.
  end if
end for.
return f_θ.

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In the end, the loss on both the source and target domains is employed to update the training process.

IV. Experiment Settings

A. Datasets

Two publicly available datasets, i.e., IsprsDA datasets [42] and LoveDA datasets [43], are employed to demonstrate the effectiveness of the proposed method.

1) IsprsDA: The dataset is collected by satellite in orthographic projection focusing on urban scenes. It contains two domains: Vaihingen and Potsdam. The size of images varies from 1996 × 1995 pixels to 3816 × 2550 pixels in Vaihingen and 6000 × 6000 in Potsdam. Furthermore, the ground sampling distance (GSD) of the datasets is 9 cm. Near-infrared (IR), red (R), green (G), and blue (B) bands are the basic spectral bands for the input images. Moreover, there are 33 labeled images in Vaihingen and 13 images in Potsdam with the same categories. The number of training, validating, and testing sets is the same with [3]. Besides, the dataset contains six categories including impervious surfaces (ImpSurf), building (Building), low vegetation (LowVege), tree (Tree), and clutter/background (Clutter).

2) LoveDA: The dataset is captured from Google Earth, which contains two domains: Urban and Rural. The image size is uniform across them, measuring 1000 × 1000 pixels. Each image contains three spectral bands, i.e., red, green, and blue bands. Urban contains 2653 labeled images where 1156 images are used for training, 677 images for validating, and 820 images for testing. Rural contains 3334 labeled images where 1366 images are for training, 992 images for validating, and 976 images for testing. To be specific, the validation set is employed to generate pseudo-labels and validate the effectiveness of the proposed model, which follows the official data processing procedure [43]. Besides, the dataset contains seven categories including background (BGround), build (Build), road (Road), water (Water), barren (Barren), forest (Forest), and Agriculture (Agric).

B. Implementation

All the experiments are conducted with a single NVIDIA RTX 3090-24GB GPU on Ubuntu 20.04 LTS OS. The proposed method is implemented based on PyTorch-1.8.1+cu111. Each experiment is repeated three times using different seeds to reduce the effect of randomness. Moreover, the listed scores represent the average value of three experiments.

In this study, we employ DeeplabV2 as the fundamental segmentation network, in which the SGD is used as an optimizer with a weight decay of 0.0005. The batch size is set to 8. The learning rate is initialized to $1e^{-2}$, and its descent strategy is a poly policy formulated as $(1 - (\text{iter}/\max_\text{iter}))^{\text{power}}$, power = 0.9. We use the warm-up strategy before the learning rate descends. Additionally, DeeplabV2 is trained with 5000 iterations when Vaihingen → Potsdam and Urban → Rural. For Potsdam → Vaihingen and Rural → Urban, the number of iterations is set to be 10 000. Several widely used data augmentation strategies, i.e., random crops, random horizontal and vertical flips, and random rotation are exploited. The cropping size is set as 512 × 512 for all domains. The softmax temperature η is set to 2.0 in (10), (12), and (16). The gated threshold u_t is set to be 0.7 in (19). α, γ, and τ_0 are set to 0.996, 0.8, and 0.6, respectively.

C. Evaluation Metric

To evaluate the effect of the proposed method, a commonly used metric, i.e., intersection over union (IoU), is utilized to assess the accuracy of a single class. When it comes to the whole accuracy across all classes, mean IoU (mIoU) is adopted. The computational procedure of IoU and mIoU can be formulated as follows:

$$\text{IoU}^{\text{c}} = \frac{\mathbf{N}^{cc}}{\sum_{c'=1}^{C} \mathbf{N}^{cc'} + \sum_{c'=1}^{C} \mathbf{N}^{c'c} - \mathbf{N}^{cc}} \times 100\% \tag{21a}$$

$$\text{mIoU} = \frac{1}{C} \sum_{c'=1}^{C} \text{IoU}^{c'} \tag{21b}$$

where $\mathbf{N}^{ij}$ indicates the number of examples that belong to class i being predicted as class j.

V. Results and Analysis

In this section, we first compare our method to other classic or recently proposed methods on several cross-domain scenes. Then, we introduce the ablation studies and analyze the effectiveness of the proposed modules along with their hyperparameters.

A. Comparisons With Existing Methods

In this section, we validate the effectiveness of the proposed method compared with existing approaches on three cross-domain remote sensing scenes, including: 1) scenes across different spatial locations, 2) scenes with varying spectral characteristics, and 3) scenes that are both crossing location and crossing spectral.

1) Experiments on Cross-Location Scene: Experiments on the Potsdam (IRRG) → Vaihingen (IRRG) and Vaihingen (IRRG) → Potsdam (IRRG) tasks are conducted first. In these experiments, several classical or recently proposed methods are adopted for comparison, including AdaptSeg [20], FADA [21], PyCDA [44], IAST [26], CutMix [27], DACS [25], DecoupleNet [19], ProCA [23], DCA [6], and MuGCDA [3]. These methods cover adversary training, ST, CL, and MIX techniques. The basic semantic segmentation network of these methods is set to DeepLabV2 with a pretrained Resnet-101 backbone. Similar to [3] and [45], the category Clutter is ignored when testing. The experiment results are listed in Tables I and II. The network trained solely on the source domain data (Source-only) gets 43.37%/39.70% mIoU scores on these tasks. The performance of the network trained by labeled data of the target domain (Oracle) achieves 68.73%/79.73% mIoU scores. Our method improves the performance of each category compared with the Source-only model, which indicates the effectiveness of the proposed method in domain adaptation. It can be seen that our method gains the best overall performance compared with other classic

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TABLE I

NUMERICAL COMPARISON WITH EXISTING METHODS ON POTSDAM (IRRG) → VAIHINGEN (IRRG) TASK. † INDICATES THAT THE RESULTS OF THE METHODS ARE REPORTED BASED ON OUR EXPERIMENTS. OTHER CITED SCORES ARE REFERRED FROM [3]

Domains	Methods	Type	IoU(%)					mIoU(%)
			ImpSurf	Building	LowVege	Tree	Car	
Potsdam (IRRG) ↓ Vaihingen (IRRG)	Source-only	-	45.73	58.57	34.70	57.91	19.96	43.37
	Oracle	-	76.25	83.32	62.35	72.36	46.35	68.73
	AdaptSeg [20]	AT	58.10	74.07	45.62	60.03	13.92	50.35
	FADA [21]	AT	60.01	70.96	47.23	58.06	26.79	52.61
	PyCDA [44]	ST	63.11	71.74	33.11	60.78	29.65	51.68
	IAST [26]	AT+ST	64.62	70.28	44.90	62.11	37.57	55.90
	DecoupleNet [19]	ST	64.49	70.45	47.42	62.12	36.27	56.15
	CutMix† [27]	MIX+ST	**70.01**	**81.03**	51.24	63.70	38.15	60.83
	DACS† [25]	MIX+ST	66.14	80.30	50.06	64.46	**41.40**	60.47
	ProCA† [23]	CL+ST	67.69	80.61	48.86	64.14	36.14	59.49
	DCA† [6]	CL+ST	68.61	77.85	50.71	65.18	39.03	60.28
	MuGCDA [3]	ST	69.02	76.42	50.24	62.55	34.29	58.50
	Ours†	ST	69.97	80.43	**53.90**	**65.63**	35.87	**61.16**

TABLE II

NUMERICAL COMPARISON WITH EXISTING METHODS ON VAIHINGEN (IRRG) → POTSDAM (IRRG) TASK. † INDICATES THAT THE RESULTS OF THE METHODS ARE REPORTED BASED ON OUR EXPERIMENTS. OTHER CITED SCORES ARE REFERRED FROM [3]

Domains	Methods	Type	IoU(%)					mIoU(%)
			ImpSurf	Building	LowVege	Tree	Car	
Vaihingen (IRRG) ↓ Potsdam (IRRG)	Source-only	-	48.06	59.29	44.56	8.98	37.09	39.70
	Oracle	-	82.46	90.80	72.27	74.07	79.05	79.73
	AdaptSeg [20]	AT	**66.36**	**75.56**	52.97	25.90	46.68	53.49
	FADA [21]	AT	61.13	62.92	48.90	33.92	41.56	49.69
	PyCDA [44]	ST	55.33	52.83	48.02	22.37	56.97	47.10
	IAST [26]	AT+ST	62.35	68.19	53.94	39.89	43.66	53.61
	DecoupleNet [19]	ST	59.10	59.15	55.10	42.09	46.39	52.37
	CutMix† [27]	MIX+ST	57.11	63.59	44.88	37.07	59.19	52.37
	DACS† [25]	MIX+ST	58.08	65.57	50.29	46.62	62.48	56.61
	ProCA† [23]	CL+ST	56.98	60.72	54.86	**49.63**	53.56	55.15
	DCA† [6]	CL+ST	49.34	56.59	42.06	26.34	58.75	46.62
	MuGCDA [3]	ST	60.32	65.36	**57.53**	43.00	52.35	55.71
	Ours†	ST	57.52	67.71	55.16	46.30	**66.61**	**58.66**

and recent proposed methods. Specifically, our method outperforms the famous AdaptSeg [20] by solid margins across mIoU (8.15%/5.17%). Furthermore, it improves the performance by 2.35%/2.95% than the recently proposed method MuGCDA [3]. Our method outperforms other methods on LowVege and Tree while getting competitive performance on ImpSurf, Building, and Car. These numerical results demonstrate the effectiveness of the proposed method.

The visualized qualitative results of the compared methods, ours, and the ground truth (GT) on Potsdam (IRRG) → Vaihingen (IRRG) task are illustrated in Fig. 5. It can be seen that the segmentation results of our method demonstrate a closer alignment with GT, exhibiting the least errors and the most reasonable visual results. Notably, Source-only, AdaptSeg, PyCDA, and FADA methods encounter obvious mix-detection between ImpSurf and Building, as well as between LowVege and Tree, due to relatively small inter-class differences. Although other compared methods achieve high scores, their visualization results display unrealistic false detection noise. In contrast, our method yields more regular and smoother building edges without substantial confusion with ImpSurf, indicating the effectiveness of our noise correction and loss reweighting approach.

We also conduct experiments on the LoveDA datasets to demonstrate the effectiveness in cross-location scenes. LoveDA contains Rural → Urban and Urban → Rural tasks. In the experiment, MCD [46], AdaptSeg [20], FADA [21], CLAN [47], TransNorm [48], PyCDA [44], CBST [10], IAST [26], CutMix [27], DACS [25], ProCA [23], and DCA [6] are selected to be the compared methods. The basic semantic segmentation network of these methods is set to DeepLabV2 with a pretrained Resnet-50 backbone, which is the same as introduced in [43]. As shown in Tables III and IV, our method outperforms the famous MCD by solid margins across mIoU (4.33%/10.41%). Furthermore, it improves the performance by 3.31%/0.93% than the current state-of-the-art method DCA. These numerical results also provide evidence for the effectiveness of the proposed approach in the cross-location scene.

2) Experiments on Cross-Spectral Scene: We conduct the experiment on Potsdam (RGB) → Potsdam (IRRG) task to evaluate the domain adaptation performance in cross-spectral

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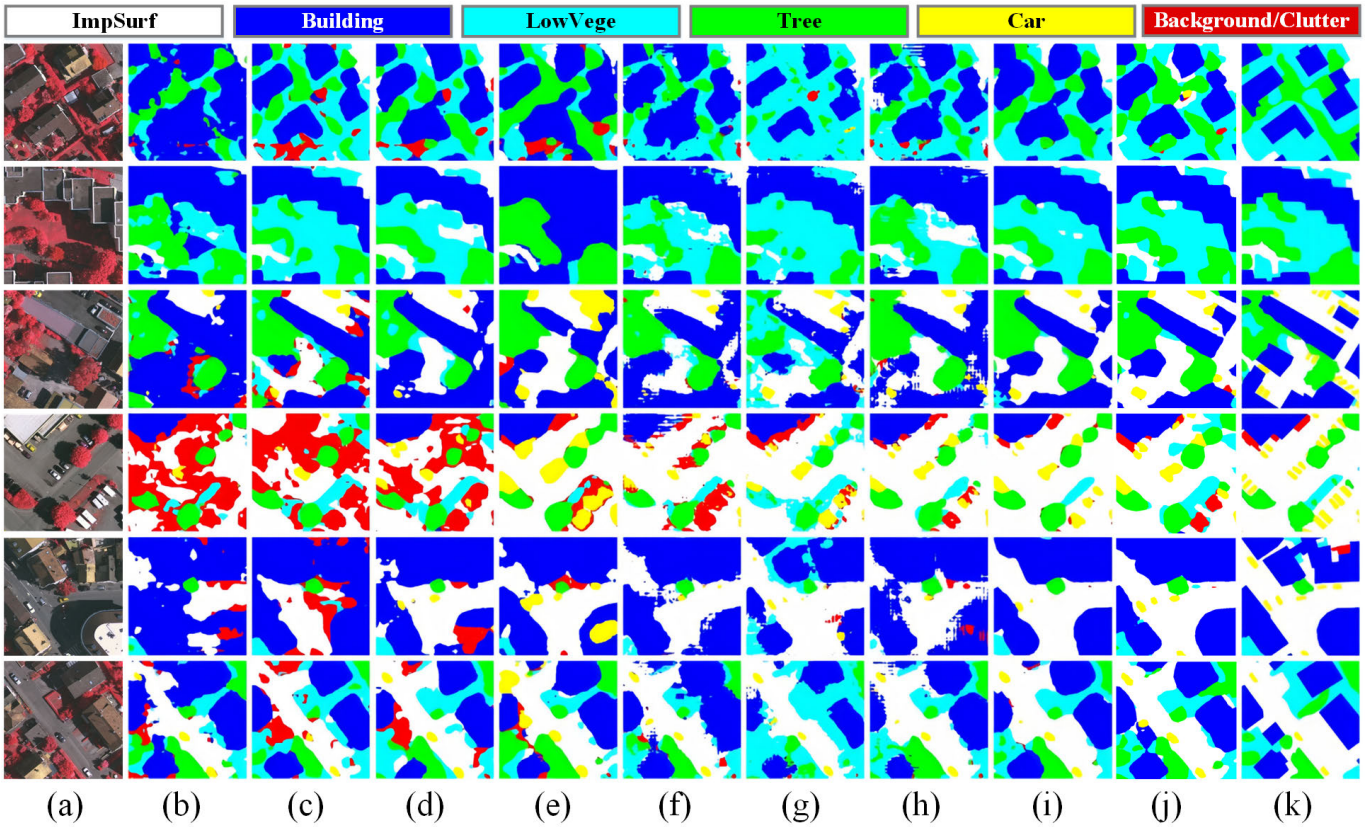


Fig. 5. Visualization results of the Potsdam (IRRG) → Vaihingen (IRRG) task. (a) Input images of Vaihingen (IRRG). (b) Results of the Source-only model. (c) Results of AdaptSeg. (d) Results of FADA. (e) Results of PyCDA. (f) Results of IAST. (g) Results of DACS. (h) Results of DecoupleNet. (i) Results of MuGCDA. (j) Results of our method. (k) Ground truth.

TABLE III

NUMERICAL COMPARISON WITH EXISTING METHODS ON THE RURAL→URBAN TASK. † INDICATES THAT THE RESULTS OF THE METHODS ARE REPORTED BASED ON OUR EXPERIMENTS. OTHER CITED SCORES ARE REFERRED FROM [43]

Domains	Methods	Type	IoU(%)							mIoU(%)
			BGround	Build	Road	Water	Barren	Forest	Agric	
Rural ↓ Urban	Source-only	-	43.30	25.63	12.70	76.22	12.52	23.34	25.14	31.27
	Oracle	-	48.18	52.14	56.81	85.72	12.34	36.70	35.66	46.79
	MCD [46]	-	43.60	15.37	11.98	79.07	14.13	33.08	23.47	31.53
	AdaptSeg [20]	AT	42.35	23.73	15.61	81.95	13.62	28.70	22.05	32.68
	FADA [21]	AT	43.89	12.62	12.76	80.37	12.70	32.76	24.79	31.41
	CLAN [47]	AT	43.41	25.42	13.75	79.25	13.71	30.44	25.80	33.11
	TransNorm [48]	AT	38.37	5.04	3.75	80.83	14.19	33.99	17.91	27.73
	PyCDA [44]	ST	38.04	35.86	45.51	74.87	7.71	40.39	11.39	36.25
	CBST [10]	ST	48.37	46.10	35.79	80.05	19.18	29.69	30.05	41.32
	IAST [26]	AT+ST	**48.57**	31.51	28.73	**86.01**	**20.29**	31.77	**36.50**	40.48
	CutMix† [27]	ST+MIX	42.96	48.27	44.66	76.69	13.52	38.81	28.09	41.86
	DACS† [25]	ST+MIX	46.68	46.52	44.94	81.90	16.42	32.13	30.14	42.68
	ProCA† [23]	ST+CL	43.53	34.58	39.54	69.97	13.96	40.91	28.78	38.75
	DCA† [6]	ST+CL	42.60	45.72	49.03	73.68	16.60	38.89	28.91	42.35
	Ours†	ST	46.91	**48.28**	**49.40**	83.89	15.58	**41.26**	34.29	**45.66**

scenes. The experiment results are listed in Table V. The scores of all the compared methods in this task are much greater than the ones in Potsdam (IRRG) → Vaihingen (IRRG) task, which indicates that the cross-location adaptation task is relatively easier than the cross-spectral (RGB to IRRG) ones. Furthermore, our method achieves the best performance on the overall score and yields 77.23% mIoU and practically catches up with the Oracle model. It can be inferred that our method has better cross-spectral adaptation than the compared methods. Moreover, although the Source-only model gains 58.51% mIoU score, the IoU score of the category LowVege is only 10.43%. This phenomenon shows that the spectral differences caused by varying image bands significantly impair the model's ability to distinguish this category. It is worth noting

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TABLE IV

NUMERICAL COMPARISON WITH EXISTING METHODS ON THE URBAN→RURAL TASK. † INDICATES THAT THE RESULTS OF THE METHODS ARE REPORTED BASED ON OUR EXPERIMENTS. OTHER CITED SCORES ARE REFERRED FROM [43]

Domains	Methods	Type	IoU(%)							mIoU(%)
			BGround	Build	Road	Water	Barren	Forest	Agric	
Urban ↓ Rural	Source-only	-	24.16	37.02	32.56	49.42	14.00	29.34	35.65	31.74
	Oracle	-	37.18	52.74	43.74	65.89	11.47	45.78	62.91	45.67
	MCD [46]	AT	25.61	44.27	31.28	44.78	13.74	33.83	25.98	31.36
	AdaptSeg [20]	AT	26.89	40.53	30.65	50.09	16.97	32.51	28.25	32.27
	FADA [21]	AT	24.39	32.97	25.61	47.59	15.34	34.35	20.29	28.65
	CLAN [47]	AT	22.93	44.78	25.99	46.81	10.54	37.21	24.45	30.39
	TransNorm [48]	AT	19.39	36.30	22.04	36.68	14.00	**40.62**	3.30	24.62
	PyCDA [44]	ST	12.36	38.11	20.45	57.16	18.32	36.71	41.90	32.14
	CBST [10]	ST	25.06	44.02	23.79	50.48	8.33	39.16	49.65	34.36
	IAST [26]	AT+ST	29.97	49.48	28.29	**64.49**	2.13	33.36	61.37	38.44
	CutMix† [27]	ST+MIX	29.13	54.07	33.52	54.91	15.95	33.82	55.08	39.50
	DACS† [25]	ST+MIX	28.11	53.43	30.22	58.72	20.20	36.00	58.60	40.75
	ProCA† [23]	ST+CL	29.93	53.62	32.22	54.77	20.04	34.50	53.21	39.76
	DCA† [6]	ST+CL	27.91	**55.42**	27.43	61.22	22.73	34.28	56.86	40.84
	Ours†	ST	**30.54**	52.42	**33.07**	61.09	**24.75**	33.67	**56.87**	**41.77**

TABLE V

NUMERICAL COMPARISON WITH EXISTING METHODS ON THE POTSDAM (RGB) → POTSDAM (IRRG) TASK. † INDICATES THAT THE RESULTS OF THE METHODS ARE REPORTED BASED ON OUR EXPERIMENTS. OTHER CITED SCORES ARE REFERRED FROM [3]

Domains	Methods	Type	IoU(%)					mIoU(%)
			ImpSurf	Building	LowVege	Tree	Car	
Potsdam (RGB) ↓ Potsdam(IRRG)	Source-only	-	72.27	75.85	10.43	66.06	66.17	58.16
	Oracle	-	82.46	90.80	72.27	74.07	79.05	79.73
	AdaptSeg [20]	AT	73.80	80.81	51.19	67.18	69.56	68.51
	FADA [21]	AT	75.91	83.97	62.06	68.77	66.83	71.51
	PyCDA [44]	ST	76.41	82.70	63.49	69.31	73.69	73.21
	IAST [26]	AT+ST	76.20	83.67	54.29	68.26	66.81	69.85
	DecoupleNet [19]	ST	76.21	82.25	59.50	68.10	72.97	71.81
	CutMix† [27]	MIX+ST	**80.01**	87.27	45.35	72.18	75.05	71.97
	DACS [25]	MIX+ST	74.09	81.14	63.44	66.83	71.16	71.33
	ProCA† [23]	CL+ST	79.69	**89.39**	65.45	71.11	**76.67**	76.46
	DCA† [6]	CL+ST	74.04	81.81	45.59	67.11	70.42	67.80
	MuGCDA [3]	ST	78.05	86.78	66.84	69.72	68.41	73.96
	Ours†	ST	79.76	88.42	**68.77**	**72.63**	76.57	**77.23**

that the performance of this category increases by 58.34% after employing the proposed domain adaptation method, which also indicates the effectiveness of the proposed method.

3) Experiments on Cross-Position and Cross-Spectral Scene: We conduct the experiment on the Potsdam (RGB) → Vaihingen (IRRG) task to evaluate the domain adaptation performance in cross-position and cross-spectral scenes. As shown in Table VI, the scores of all the compared methods in this task are much smaller than the ones in Potsdam (IRRG) → Vaihingen (IRRG) and Potsdam (RGB) → Potsdam (IRRG) tasks, which indicates that the cross-location and cross-spectral task is relatively harder than the cross-location task or cross-spectral task. Our method achieves greater mIoU scores than the compared methods. Notably, our method outperforms the Source-only model on all categories, which indicates that our method works on all categories. Comparing the recently proposed MuGCDA, our method can gain a better score on Building category by a large margin and the IoU scores of other categories are similar, which results in our outstanding performance over all categories.

B. Ablation Study

1) Influence of the MPC and the UVEM: In this section, the effectiveness of the designed MPC and the UVEM modules is discussed. The results are summarized in Table VII. The second row refers to the baseline method that is detailed in Section III-B. Three crucial conclusions can be drawn from the results. First, our MPC leads to significant improvements against the baseline on both Vaihingen (IRRG) → Potsdam (IRRG) and Potsdam (IRRG) → Vaihingen (IRRG) tasks. This illustrates that our MPC module can well correct the pseudo-labels gaining better segmentation performance in target domains. Second, compelling gains are obtained by incorporating the UVEM algorithm into the baseline. This proves the importance of reweighting for the examples with different uncertainty. Third, our model achieves the best performance by jointly utilizing the proposed MPC and UVEM

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TABLE VI

NUMERICAL COMPARISON WITH EXISTING METHODS ON THE POTSDAM (RGB) → VAIHINGEN (IRRG) TASK. † INDICATES THAT THE RESULTS OF THE METHODS ARE REPORTED BASED ON OUR EXPERIMENTS. OTHER CITED SCORES ARE REFERRED FROM [3]

Domains	Methods	Type	IoU(%)					mIoU(%)
			ImpSurf	Building	LowVege	Tree	Car	
Potsdam (RGB) ↓ Vaihingen (IRRG)	Source-only	-	36.68	52.15	2.46	48.04	24.52	32.77
	Oracle	-	79.36	85.77	64.64	74.32	59.11	72.64
	AdaptSeg [20]	AT	48.21	60.45	16.45	57.60	17.79	40.10
	FADA [21]	AT	56.66	65.89	**36.34**	34.39	27.36	44.13
	PyCDA [44]	ST	56.80	72.67	17.10	56.01	23.79	45.27
	IAST [26]	AT+ST	52.06	69.27	6.74	59.55	31.46	43.82
	DecoupleNet [19]	ST	53.06	60.72	16.82	60.60	33.48	44.94
	CutMix† [27]	MIX+ST	54.15	68.49	15.44	58.71	32.80	45.92
	DACS [25]	MIX+ST	61.89	61.24	35.89	5.58	**48.24**	42.57
	ProCA† [23]	CL+ST	58.56	73.13	16.40	56.98	30.52	47.12
	DCA† [6]	CL+ST	61.98	68.01	11.91	60.21	35.89	47.60
	MuGCDA [3]	ST	**63.70**	53.53	11.79	**63.59**	37.59	46.04
	Ours†	ST	60.39	**75.06**	19.62	58.67	37.84	**50.32**

TABLE VII

ANALYSIS OF THE ESSENTIAL COMPONENTS ON THE ISPRSDA DATASETS

Methods	Vaihingen→Potsdam	Potsdam→Vaihingen
Source-only	39.70	43.37
Baseline	52.18	58.66
Base+MPC	55.76	59.66
Base+UVEM	53.87	60.20
ours	**58.66**	**61.16**

TABLE VIII

ANALYSIS OF THE EFFECTIVENESS OF MULTIPLE LEVELS IN MPC WHEN ADAPTING FROM VAIHINGEN (IRRG) TO POTSDAM (IRRG)

Stages	Components				mIoU
stage 1	Source-only				46.09
	self-training	class-level	pixel-level	superpixel-level	
stage 2	✓				52.18
	✓	✓			53.69
	✓		✓		53.08
	✓			✓	52.87
	✓	✓	✓	✓	**55.76**

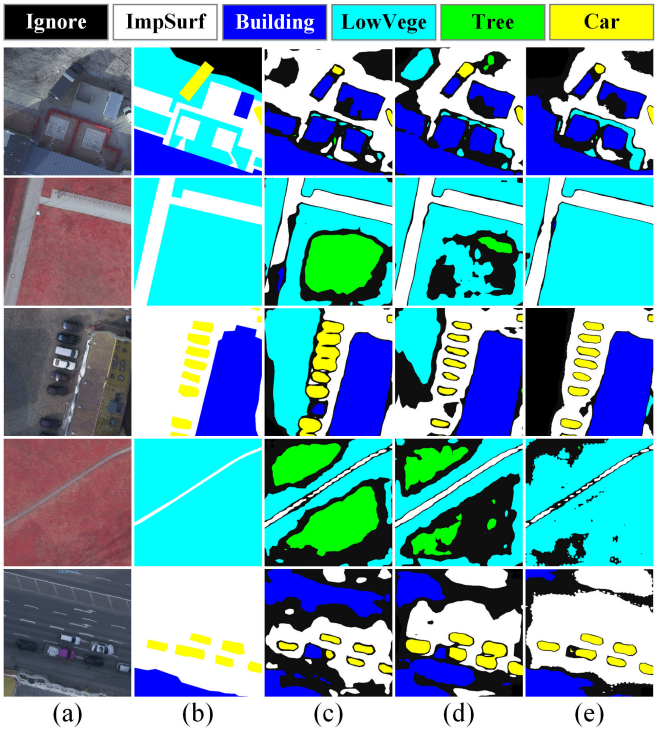


Fig. 6. Visualization of the pseudo-labels generated by compared methods on Vaihingen (IRRG) → Potsdam (IRRG) task. (a) Input image. (b) Ground truth. (c) Noisy pseudo-labels were generated by the baseline method without correcting. (d) Pseudo-labels generated by ProDA [8] which corrects the pseudo-labels in class-level. (e) Pseudo-labels generated by the proposed MPC. Examples within the black area are not selected to be pseudo-labels.

modules, confirming the necessity of both the proposed MPC module and the UVEM algorithm.

2) Effectiveness of the Levels in MPC: . The effectiveness of the proposed MPC methods is analyzed. The experiment results from Vaihingen (IRRG) to Potsdam (IRRG) are listed in Table VIII. After the utilization of the correcting weights from the class-level, pixel-level, or superpixel-level, the proposed framework can obtain considerable gains. It indicates that the local consistency of images, compactness of feature space, and the online predictions generated by the student model can help correct the pseudo-labels. Furthermore, when multi-level correcting strategies are jointly employed, the proposed method achieves the highest mIoU score. Moreover, the visual comparison of the pseudo-labels is shown in Fig. 6. The pseudo-labels corrected by the MPC module are well aligned with the ground truth compared to other methods. These numerical and visual results strongly prove that the MPC module makes a great contribution to pseudo-label correcting by taking full advantage of the complementary information in the original image, deep features, and real-time prediction.

3) Effectiveness of the UVEM: To demonstrate the effectiveness of the UVEM, several example mining methods are used for comparison. As shown in Table IX, our UVEM strategy outperforms other example mining methods. The loss-based OHEM and Focal methods get a worse performance

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TABLE IX

COMPARISON WITH OTHER EXAMPLE MINING METHODS ON THE CROSS-LOCATION SCENE. THE SPECTRAL OF THE DOMAINS IS IRRG

Methods	Vaihingen→Potsdam	Potsdam→Vaihingen
Baseline(CE)	52.18	58.66
OHEM [37]	49.22	58.70
Focal [38]	49.15	57.84
GHM [14]	50.32	56.03
UPS [13]	52.56	59.38
UVEM	**53.87**	**60.20**

TABLE X

EXPERIMENTS RESULTS TO DETERMINE THE SUPERPIXEL NUMBER/SIZE IN LSC ON POTSDAM (IRRG) → VAIHINGEN (IRRG)

Number	262144	16384	4096	1024	256	64
Size	1×1	4×4	8×8	16×16	32×32	64×64
mIoU	58.66	58.70	59.06	**59.12**	58.65	58.12

TABLE XI

EXPERIMENTS RESULTS ON VALIDATING SET TO DETERMINE THE OPTIMAL VALUE FOR THE UNCERTAINTY THRESHOLD u_m FROM POTSDAM (IRRG) TO VAIHINGEN (IRRG) SET WHEN u_t IS SET TO 0.7 AND ρ IS SET TO 1.0

u_m	0	0.1	0.2	0.3	0.4	0.5
mIoU	59.44	59.68	**60.54**	60.16	60.14	59.79

than the baseline method (cross-entropy loss). This result indicates that focusing on hard examples of the target domain has side effects on the model convergence. The reason is that hard examples have a high noise rate leading to a quick error spreading on the ST process. GHM also performs worse than the baseline method due to the sparsity of the gradient density for hard examples with high noise rates. UPS slightly outperforms the baseline method by filtering out examples with a high likelihood of being false. By contrast, our UVEM performs better than UPS due to its effectiveness in mining valuable examples.

4) Influence of the Superpixel Number/Size in MPC: We also study the effect of the superpixel number/size. The mIoU score vs superpixel number/size is listed in Table X. It is evident that limited improvement is gained when setting the superpixel size to be small. The optimal performance is obtained when the size is 16 × 16. However, as the superpixel size increases, the performance deteriorates since the generated larger superpixels may not adhere to local consistency.

5) Influence of the Uncertainty Threshold u_m: To determine the optimal uncertainty threshold u_m, an experiment is conducted on Potsdam (IRRG) → Vaihingen (IRRG) task. As depicted in Table XI, the segmentation result obtained by the proposed method initially increases with the increment of u_m and subsequently exhibits a decreasing trend. Moreover, the proposed model achieves satisfactory performance when the uncertainty threshold u_m is set to a moderate value (0.2, 0.3, or 0.4). When the uncertainty threshold u_m is set to 0 or 0.7 (the same as u_t), the proposed model gains poor performance. This experiment suggests that the most beneficial examples for ST are those with moderate uncertainty rather than border uncertainty. Moreover, the experimental result demonstrates that both easy examples with limited gradients and hard examples with high errors should be suppressed in the ST stage.

TABLE XII

EXPERIMENTS RESULTS ON VALIDATING SET TO DETERMINE THE OPTIMAL VALUE FOR THE SMOOTH ADJUSTING FACTOR ρ FROM POTSDAM (IRRG) TO VAIHINGEN (IRRG) WHEN u_m IS SET TO 0.2

ρ	0.5	1	2	4	8	16	32
mIoU	60.42	60.54	**60.72**	60.33	60.43	60.20	60.18

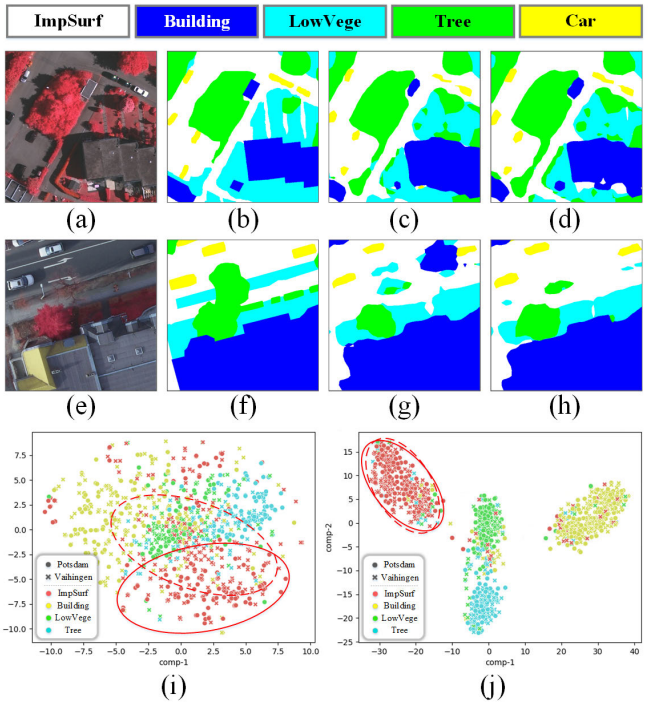


Fig. 7. Illustration of the domain adaptive effectiveness of the proposed method on Vaihingen (IRRG) → Potsdam(IRRG) task. (a) Source image of Vaihingen (IRRG). (b) Ground truth of the source image. (c) Result of the Source-only model. (d) Result of our model. (e) Target image of Potsdam (IRRG). (f) Ground truth of the target image. (g) Result of the Source-only model. (h) Result of our model. (i) and (j) 2-D t-SNE feature distribution yield by the Source-only model and our model, respectively.

6) Influence of the Adjusting Factor ρ: To investigate the impact of adjusting factor ρ on the UVEM performance, an experiment is conducted on the Potsdam (IRRG) → Vaihingen (IRRG) task. Table XII lists the model performance over different ρ values when $u_m = 0.2$ and $u_t = 0.7$. It is shown that the segmentation performance of the proposed method tends to decrease as ρ increases. The reason is that an excessively large ρ makes the valuable weight curve too plant to focus on mining the valuable examples. The performance waves slightly when ρ ranges from 0.5 to 8, which indicates the proposed UVEM is not sensitive to the adjusting factor ρ.

7) Effectiveness of the Proposed Method: To illustrate the effectiveness of the proposed domain adaptive method, a visual comparison is illustrated in Fig. 7. We can see that both the Source-only model and the proposed one can yield considerable results in the source domain. When testing in the target domain, the performance of the Source-only model decreases a lot, while the proposed method can still yield satisfactory results. Furthermore, the distributions of deep features are plotted to demonstrate the domain adaptive ability of the

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proposed method. As shown in Fig. 7(i), the deep features learned by the Source-only model between the source domain and target domain are separate because of the domain gap. As shown in Fig. 7(j), it can be seen that our method can explicitly learn the shared representations of the same categories from different domains. Besides, our method can drive the intracategory features closer and intercategory features further apart, which builds more accurate segmentation results on the target domain.

C. Discussion

The experiments conducted on the cross-location and cross-spectral datasets demonstrate the effectiveness of the proposed method. However, the ability of our method to handle scenarios like cross-period (e.g., summer → winter or day → night) and significantly different sensor types (e.g., far infrared → RGB) remains unexplored. Besides, dealing with inconsistent category spaces between the source and target domains is still a challenging issue.

VI. Conclusion

In this work, we present an ST-based approach to address the UDA problem for RSIS. We propose two novel modules to tackle the issue of noisy pseudo-labels, a key challenge in ST-based UDA: MPC and UVEM. On the one hand, MPC adjusts the soft pseudo-labels online in class, pixel, and superpixel levels. By incorporating it, more reliable pseudo-labels can be selected for subsequent training stages. On the other hand, UVEM assigns low weight to noisy pseudo-labels when computing the loss in the target domain. It suppresses both easy examples with limited gradients and hard examples with high error rates based on example uncertainty. By incorporating UVEM, more confident examples with significant gradients can be prioritized for training, which makes the model convergence better. Empirical evaluations on IsprsDA and LoveDA datasets demonstrate that the proposed approach outperforms previous state-of-the-art methods in the cross-location and cross-spectral domain adaptation tasks.

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